

Jack Wong

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Education

Carleton University | *B.Eng. in Computer Systems Engineering*

Exp. 2029

GPA: 3.95/4.0

- Activities & Societies: SCESoc VP Services, Leo's Lounge Shift Volunteer, EngFrosh Facilitator

Experience

Software Developer | *Carleton Planetary Robotics Team (CPRT) – Ottawa, ON*

Sept 2025 – Present

- Developed Node.js service to track device presence and RTT ping times through periodic ping measurements, providing real-time visibility in a React/TypeScript UI.
- Reduced post-processing latency by 15 ms and improved real-time path planning performance by implementing a lookup table for gridmap inflation decay values, to reduce CPU usage.
- Achieved a 180 degree turn in under 1 second by tuning PID control parameters for swerve drive motor.
- Presented technical improvements and performance results to professors and industry alumni during the Project Design Review (PDR) as part of preparation for the University Rover Competition (URC).

Software & Outreach Member | *Spark Youth Robotics Club (FRC 8729) – Kanata, ON*

Sept 2022 – Aug 2025

- Organized and ran STEM workshops for intermediate students at 5 schools across Ottawa, focused on circuits and building presentation skills.
- Participated in the Robot Demo event at Carlingwood Mall, helping to increase community outreach and support from across Ottawa by showcasing the team's accomplishments.
- Collaborated with team members to code 3 multi-mechanism robots in Java using the custom WPILib library.
- Demonstrated creativity and problem solving while competing in 8 FIRST Robotics competitions, including all 3 provincial championships.

Teaching Assistant | *Bell High School – Ottawa, ON*

Jan 2025 – June 2025

- Mentored 30 students in a computer science class by creating individual plans with host teacher to fit specific needs.
- Created and presented lectures on logical thinking, debugging, and number representation.
- Recorded educational videos explaining topics that students were struggling with. The final exam saw multiple students improve their overall grade by 5%.

Projects

Facebook Scraper

Sept 2025 - Nov 2025

- Developed a Facebook group scraper bot using Selenium and discord.py to automatically detect posts containing configured keywords.
- Implemented real-time notifications to Discord channels for matched posts, enabling automated monitoring of social media content.
- Designed the system with randomized polling intervals and configurable settings to improve efficiency and reduce detection risk.

Paragon (demo)

Jan 2024 - July 2024

- Developed an educational game using Pygame in Python, practicing object-oriented programming, organized file handling, and version control with GitHub
- Demonstrated extensive independent problem solving and goal management capabilities

Fashion Prediction Model

July 2023 - Aug 2023

- Deep learning model trained on the Fashion-MNIST dataset to classify images of clothing into categories such as shirts, shoes, and bags.
- Model was coded and trained in Python using libraries such as TensorFlow and Keras, leveraging their powerful tools for neural network design and training.